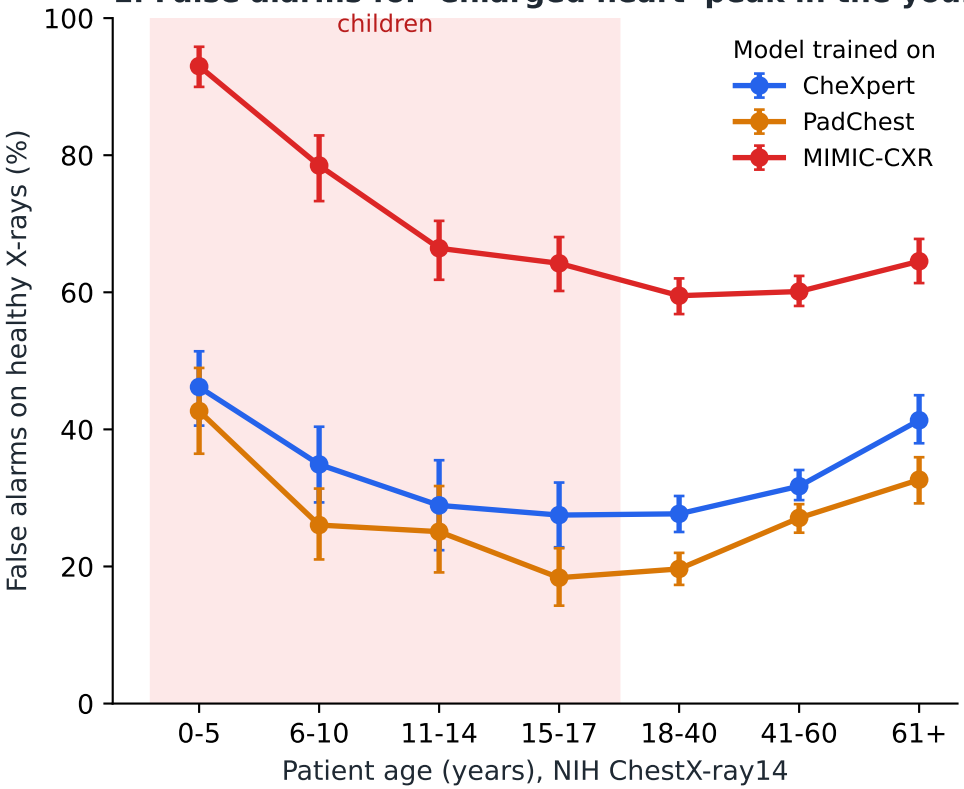
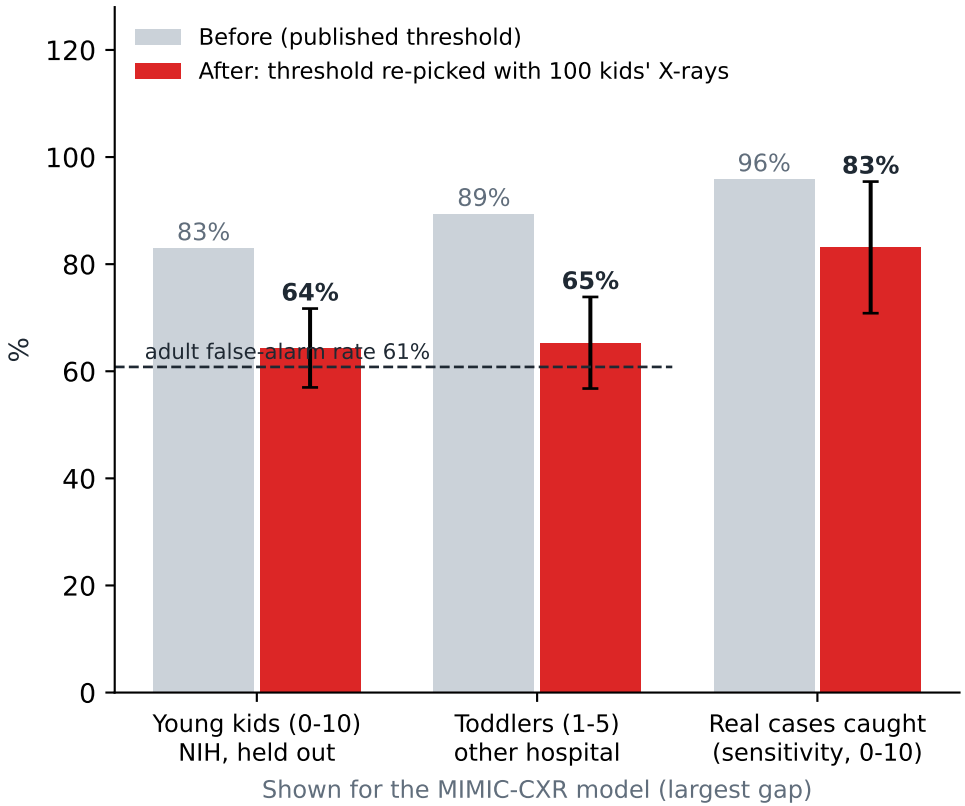
Adult-trained chest X-ray AI over-flags young children. Can a cheap fix help?

Follow-up to Hua et al., "Lack of children in public medical imaging data" (2025) · Shreya Pawar, UC Berkeley · public data and models

1. False alarms for 'enlarged heart' peak in the youngest kids



2. Fix: re-pick the threshold with 100 kids' X-rays



Shown for the MIMIC-CXR model (largest gap)

What I found Ages 0-5 get the most false 'enlarged heart' calls on healthy X-rays: CheXpert 46% vs 28% for ages 18-40; PadChest 43% vs 20% for ages 18-40; MIMIC-CXR 93% vs 60% for ages 18-40. This reproduces the paper's age-bias finding using only public data.

What helps Re-picking the decision threshold with ~100 healthy young-child X-rays (no retraining) brings the MIMIC model's young-child false alarms from 83% to 64% (adults: 61%), and the same threshold carries over to toddlers from a hospital it never saw (89% to 65%). Gains are smaller for CheXpert (39% to 34%) and PadChest (31% to 29%).

The catch Fewer false alarms cost sensitivity (96% to 83% for MIMIC; only 53 young-child cardiomegaly cases, so this is noisy). Retraining the model's last layer on NIH children did not carry over to the other hospital. Fixing the gap without missing real cases likely needs real pediatric training data, which supports the paper's call to release it.

Next Repeat on VinDr-PCXR (the paper's pediatric set) with age-specific thresholds and full fine-tuning; measure how many pediatric images it takes to recover sensitivity.

Data: NIH ChestX-ray14 (16,291 images incl. all 5,402 pediatric), Guangzhou pediatric CXR (Kermany 2018, ages 1-5).
Models: TorchXRayVision DenseNet-121 (none trained on NIH). 10 patient-disjoint splits. Code: see repo link.